

12.4 Discounts and Markups

Lesson Introduction

 Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.		 Learning Target	I can solve problems involving discounts and markups.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.		N/A	
 Essential Questions	- What are the similarities and differences between calculating discounts and markups? - How do you solve real-world problems involving discounts and markups?			
 Lesson Narrative	In this lesson, students solve problems involving discounts and markups. The lesson begins by exploring discounts and markups with familiar contexts to access students' prior knowledge. Students then work with a partner to solve problems they select from a menu of options. The lesson concludes with a student reflection that can be used to plan for upcoming lessons, as students continue working with percentages.			
 Component Summary	12.4.1 Warm-Up (5-8 min.)	Reflect on how civil engineers use math (SE p. 229)	12.4.3 Activity (15-20 min.)	Collaboratively select and solve real-world discount and markup problems (SE p. 232)
	12.4.2 Guided Instruction (15 min.)	Explore how to determine discounts and markups in context (SE p. 229)	12.4.4 Wrap-Up (~5 min.)	Reflect on which parts of the lesson were difficult or confusing (SE p. 234)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Discount <u>Additional Terms:</u> - Markup		Career Profile video	- Consider bringing in receipts and sales flyers to show students the utility of the mathematics in the real-world. 12.4.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may confuse the discount or markup amount, original cost, and final cost of a product when determining how to solve a problem.	- Facilitate a conversation with students about the differences between the discount/markup amount, original cost, and the final cost of a product. Model how to approach problems if needed and encourage students to mark up the text as they read and make sense of each problem. - During the 12.4.2 Guided Instruction, consider creating and displaying an anchor chart with instructions, examples, and visuals for students to reference that highlights different approaches to solving problems.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 12.4.3 Activity, ask students to take turns selecting menu items before they work together to solve the problems and write their work in the tables. The student who selects the problem should begin the discussion about how to approach the problem. Discussing the problems with partners will build student confidence and understanding. It also holds both students accountable.	MA.K12.MTR.5.1: Patterns and Structure In 12.4.3 Activity, students solve problems with discounts and markups. Students can use a variety of strategies. Ask students to use their reasoning and sense-making skills to determine what the problem is asking them to find and what information is provided and develop a plan to solve. Prompt students to look for patterns and structure among the discount and markup problems they solve.
 Differentiate Instruction	<i>For 12.4.2 Guided Instruction, consider providing visual and kinesthetic entry points by having students look at sales flyers to see real-world examples of discounts.</i> <i>Discussions during 12.4.3 Activity provide auditory entry points. Consider previewing the problems and having an eye on which students are selecting from the menu so that you can help push students' thinking as appropriate.</i>	
Lesson Notes:		

12.4.1 Warm-Up: Civil Engineer

Component Overview: Play the Career Profile video on the civil engineer for the entire class. After students watch the video (2:49 minutes), prompt them to reflect on the profession by answering the two questions independently.



12.4.1 Warm-Up: Civil Engineer

Alishia Ballard works as a civil engineer for San Francisco Public Works. Civil engineers work on the plans to create structures such as bridges, large buildings, etc. They also analyze whether or not structures are safe.

1. How do you think civil engineers use math in their tasks?

Answers will vary. Civil Engineers use math to calculate how much material will be needed to create a structure. Civil engineers use math to figure out what dimensions are needed to make sure the structure does not collapse.

Alishia enjoys working at Public Works because it's a way to give back to her community. Every structure plan she works on or structure she inspects is all for the people around her.

2. What is something you have done that gives back to your community, your school, or your peers?

Answers will vary.



Do you know someone who is a civil engineer or works in public works? Consider having a guest speaker join your class (virtually or in person). If time permits, consider having students return to this lesson for a project on the careers explored throughout the year.

12.4.2 Guided Instruction: Discounts and Markup

Component Overview: Students are guided through a series of questions, fill-in-the-blank statements, and problems in real-world contexts involving discounts and markups. The questions and problems provide students with strategies that can be used to determine discounts and markups by accessing students' prior experiences and without using a formula.



12.4.2 Guided Instruction: Discounts and Markup

Percentages and ratios are often used to express or solve problems involving changes in the price of goods and services.

1. The price of a shirt is \$18.50. Natalie has a coupon that reads "discount $\frac{1}{2}$ off."
 - a. With the coupon, will Natalie pay more or pay less than \$18.50 for the shirt?
Less because a coupon lowers the price.
 - b. Discuss with your partner what " $\frac{1}{2}$ off" means. Summarize your discussion.
Half off means to take away half of the price.

A **discount** is a reduction of a price for goods or services.

- c. By what percentage will the coupon reduce the price?
50%
- d. What is the discount amount, in dollars, taken off the shirt price?
\$9.25
- e. What is the discounted price, in dollars, of the shirt after the coupon is applied?
\$9.25



Students are introduced to the concept of discounts. Consider asking students to write the definition and examples in their notes or in an interactive journal. Also, facilitate a discussion about situations where discounts and markups are used in the real-world. Help students identify keywords or phrases that they can use to determine whether an item is being discounted or marked up.



- What percentage is equivalent to $\frac{1}{2}$ off?
- How can converting between equivalent forms of rational numbers help you solve discount and markup problems?

12.4.2 Guided Instruction: Discounts and Markup (continued)

2. Cassidy, a young inventor of Baby Toon, appeared on Shark Tank in 2019. Cassidy shared that it costs her \$6.60 to produce one of her baby spoons and that she sells them to customers marked up to \$15.00.

- a. Does a markup increase or decrease the original price?

Increase

- b. Discuss with your partner what you think markup means. Summarize the discussion by writing a definition for markup.

Answers will vary. The definition of markup is the increase of a number by a certain percentage. A markup is when a new number is created by adding a percentage of the original number to the original number.

- c. Complete the statements.

A markup is applied to goods or services so that the

seller is able to make a ☐ loss.
☐ profit. The markup

☐ increases
☐ decreases

the price from the cost to the seller and

is usually determined using percentages.

- d. If Cassidy placed a 100% markup on her spoons that only cost \$6.60 to produce, what would the price of a spoon be for customers?

\$13.20

- e. Discuss with your partner how you determined a 100% markup of \$6.60. After your discussion, make any revisions to part b, if necessary.

100% of \$6.60 is \$6.60. When the percentage amount is added to the original number, the sum is \$13.20.



Consider providing supports and different entry points for students struggling with the concepts of discounts and markups. For example:

- Dollar and coin manipulative tools can be used to provide visual and kinesthetic entry points for problems involving money (question 2).
- Provide an auditory entry point by allowing students to work with a partner or by modeling through a Think Aloud.

12.4.2 Guided Instruction: Discounts and Markup (continued)

- f. Jovi and Malik used two different methods to find the price of the spoon if the markup used is 61%. Part of their work is shown.

Jovi's Work	Malik's Work
$\frac{m}{6.60} = \frac{61}{100}$ $100 \cdot \frac{m}{6.60} = \frac{61}{100} \cdot 100 \cdot 6.60$ $100m = 402.6$ $m = 4.026$	$0.61(6.60) = m$ $4.026 = m$

Complete the statements.

Jovi used a percent ☐ equation ☒ proportion to find the value

of m while Malik used a percent ☐ equation. ☒ proportion. Both

methods resulted in the same value of m , which

represents the ☐ dollar amount of the markup. ☒ price of the spoon after markup.

At a 61% markup, the spoon would sell for ☐ \$4.02. ☐ \$4.03. ☐ \$10.62. ☒ \$10.63.

- g. Jovi and Malik both found the price of the spoon after a 61% markup by calculating the amount of the markup and adding it to the cost of the spoon. What percentage could they have used if they wanted to calculate the price after markup in one step?
161%



Consider asking students the following:

- What is the same and different about Jovi's and Malik's work?
- What does the value they calculated tell you about the markup and final cost of the spoon?



Challenge students to write a numeric expression representing the cost of the spoon after a 61% markup.

$6.6 + 0.61(6.6)$ or $1.61(6.6)$

12.4.3 Activity: Math Menu

Component Overview: Students work with a partner to solve real-world discount and markup problems. Students are allowed to choose their preferred problems from the categories in a math menu.



12.4.3 Activity: Math Menu

With your partner, choose your meal from the math menu shown. Select one appetizer, one main course, two side dishes, and one dessert. Put a * next to your selections. Then, solve each problem selected.

Appetizers
A1. A book has been discounted by $\frac{3}{10}$. The original price of the book is \$34.99. What is the amount being discounted from the book?
A2. Westside Skate Shop is having a storewide $\frac{1}{5}$ off sale. Find the amount saved on a \$66 skateboard.
A3. Jennea bought a scratched iPhone because it was discounted $\frac{9}{20}$ off the original price of \$795. What was the reduced price, in dollars?
Main Courses
M1. Ramin's Discount Automobiles buys a used car for \$3,000. They mark it up 65% before selling it to a customer. What was the sale price of the car?
M2. Elliott sells grab-and-go lunches. He spends \$2.70 per lunch on ingredients and then adds a 110% markup to sell. What is the selling price of one sandwich?
M3. It costs JS Surfboards \$225.00 in labor to make one Model S surfboard. The company marks up the surfboard 51% to sell. What amount, in dollars, are Model S surfboards sold for?



Take Turns

Ask students to take turns selecting menu items before working together to solve the problems and writing the work in the tables. The student who selects the problem should begin the discussion of how to approach it. Discussing the problems with partners will build student confidence and understanding. It also holds both students accountable.

12.4.3 Activity: Math Menu (continued)

Side Dishes

- S1. A concert ticket is on sale for \$129. The sign says 14% off the original price. What was the original price of the ticket?
- S2. A scientific calculator is 15% off. The sale price of the calculator is \$13.99. What is the original cost of the calculator?
- S3. The price of an apple is \$1.25. If you get a 10% discount, how much do you have to pay?
- S4. Teena bought a bicycle for \$528 after getting a discount of 12% taken off its normal price. Find the normal price of the bicycle.

Desserts

- D1. A store sells notebooks for \$1.60 after a $\frac{3}{10}$ markup. How much is the markup?
- D2. Burgermonster sells its burgers for \$1.84. To determine the selling price, the restaurant added a markup of $\frac{23}{10}$ to the cost of the ingredients used to make a single burger. What is the cost Burgermonster pays for ingredients?



- How can you determine whether the problem involves a discount or markup?
- What are the similarities and differences between the four choices for side dishes?
- Which problem would you prefer to solve? Why?



Some students may struggle completing the number of questions in this activity during one class session. Consider decreasing the number of side dishes to one question if needed.

12.4.3 Activity: Math Menu (continued)

Appetizer Selection Work and Answer

A1: $(34.99) \frac{3}{10} = \$10.50$ is being discounted from the book

A2: $\frac{1}{5}(66) = \$13.20$ is being saved on the skateboard

A3: $\frac{11}{20}(795) = \$437.25$ was the reduced price

Main Course Selection Work and Answer

M1: $3000(1.65) = \$4,950.00$ is the sale price of the car

M2: $2.70 + (1.10)(2.70) = \$5.67$ is the selling price of a sandwich

M3: $225(1.51) = \$339.75$ is the sale price of surfboards

Side Dish Selection Work and Answers

S1: $\frac{86}{100} = \frac{129}{x}$; $x = \$150.00$ was the original price of the ticket

S2: $\frac{85}{100} = \frac{13.99}{x}$; $x = \$16.46$ was the original cost of the calculator

S3: $\frac{9}{10}(1.25) = \$1.13$ is the price you pay for the apple

S4: $\frac{88}{100} = \frac{528}{x}$; $x = \$600.00$ is the normal price of the bicycle

Dessert Selection Work and Answers

D1: $\frac{1.3}{1} = \frac{1.6}{x}$; $x \approx \$1.23$
 $1.60 - 1.23 = \$0.37$ is the markup

D2: $1.84 = 3.3x$; $x = \$0.56$ is the cost of the ingredients



Students can solve the problems in different ways using what they have learned about percentages. The sample work shown in the key is not the only strategy.



Ask students that finish early to create their own discount or markup problem based on a real-world situation. Bring in receipts and sales flyers that can serve as an inspiration for students' problems.

12.4.4 Wrap-Up: Reflection

Component Overview: Students reflect on what may have been difficult or confusing as they worked with discount and markup problems. Consider student responses when planning for the next lesson.



12.4.4 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on discounts and markup?



Answers will vary.



For students who struggle expressing themselves in written form, consider alternative methods, such as:

- Using a rating system (i.e. 1=no confusion, 2=some confusion, 3=a lot of confusion) to label the questions from the lesson
- Writing a list of confusing topics
- Circling questions they found difficult in their book